

Stress-Testing Explainability in Multimodal LLMs via Prompt Ablation: High-Recall Structured Literature Reconnaissance

Executive summary

This reconnaissance targets a specific stress-test problem: **how prompt interventions (wording, order, specificity, structured formats, chain-of-thought, decomposition, role/system prompting, token-level edits, etc.) affect both (i) MLLM outputs and (ii) the quality/stability/faithfulness/plausibility of token-level visual explanations**, with an emphasis on **TAM-style token activation maps**. The core takeaway from the recent literature is that **prompt sensitivity is real (and sometimes large) in multimodal settings**, and **explanations for autoregressive MLLMs are uniquely vulnerable to “context interference”**—i.e., earlier prompt/answer tokens can contaminate token-level visual attributions for later tokens. This is the central motivation behind **Token Activation Map (TAM)**.

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A second key takeaway is that several lines of work are converging on a similar phenomenon from different angles: (a) **prompt brittleness / robustness benchmarks** for VLMs/MLLMs (e.g., PARC, “How Sensitive Are LMMs to Prompts?”, “leading questions” benchmarks), (b) **mechanistic/attribution frameworks** for MLLMs’ token generation (e.g., EAGLE; intra-modal interaction methods like MSEA/ARC), and (c) **grounding/hallucination analyses** showing that *prompt form and “cognitive” prompting* can change failure modes and the relevance of visual evidence.

2

For your prompt-ablation workstream, the literature suggests a **two-axis ablation design** is the safest and most diagnostic:

1) **Prompt factors** (semantics-preserving vs semantics-changing, system vs user, formatting, CoT, role, etc.), and

2) **Explanation factors** (token target selection, context-window policy, aggregation across heads/layers/scales, and explanation evaluation: plausibility vs faithfulness, plus stability).

Notably, **TAM explicitly argues plausibility is better suited than perturbation-based faithfulness tests for MLLMs**, because perturbations can cause inconsistent text generation and invalidate standard faithfulness protocols; it formalizes plausibility metrics such as **Obj-IoU, Func-IoU, and F1-IoU**.

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Research methodology and scope

This is a **high-recall** reconnaissance rather than a narrow systematic review: the goal is to assemble a broad, structured corpus for later screening, prioritizing **2023–2026** while retaining older foundations where they directly inform *prompt ablation* and *explanation robustness*. The result is organized as:

- **Directly relevant:** MLLM interpretability methods that produce token-level or token-conditioned visual grounding/explanations (TAM-like; attribution for autoregressive decoding).
- **Prompt sensitivity & robustness:** papers/benchmarks that quantify prompt variance or misleading prompt effects in multimodal settings (and LLM-only meta-evaluations that transfer methodologically).

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- **Prompting methods that change reasoning/grounding:** multimodal CoT, compositional CoT, visual prompting/control, prompt interaction tooling. ⁶
- **Benchmarks/datasets/toolkits and surveys:** resources enabling systematic ablations and reproducible evaluation. ⁷

Because you specifically prefer **Qwen2-VL, LLaVA-1.5, InternVL2.5** as targets, this corpus also includes papers that explicitly evaluate or discuss those families (e.g., TAM evaluates across Qwen2-VL/ InternVL2.5/LLaVA-1.5; PARC reports InternVL2-family reliability in prompt-variation analysis; several robustness benchmarks include these families). ⁸

Annotated corpus for screening

Each entry includes: **title; year; first author; venue/source type; category; short summary; why relevant; informs output/explainability/both; target model family; code; confidence; flags.** “Peer-reviewed” means major conferences/journals; “preprint” includes arXiv/OpenReview under review; “non-academic” are blogs/repos/issues and are listed separately at the end.

Comparative table for quick triage

Relevance score is a **screening heuristic (1–5)**: 5 = directly on MLLM explainability + prompt/context effects; 4 = strong for prompt robustness or controllable prompting that affects grounding; 3 = methodologically useful (LLM robustness, CLIP/Vit interpretability); 2–1 = peripheral.

Paper (short title)	Year	Category	Tasks/settings	Model focus	Prompt intervention types studied	Key findings on prompt-ablation X explainability
Token Activation Map (TAM)	2025	MLLM+explainability (direct)	captioning, VQA, video, multi-turn	Qwen2-VL / InternVL2.5 / LLaVA-1.5	(implicit) context-token effects; evaluation relies on prompt tokens	Context tokens interfere w later-token explanation; proposes causal mitigation plausibility metrics (O Func/F1-Ic
PARC (Prompt sensitivity framework)	2025	MLLM+prompt robustness (direct)	MC-VQA	many VLM families incl. InternVL2	reformulations vs semantic changes (incl. negation), language+vision prompt variations	Realistic prompt variations calibrated reliability score; semantic changes (negations) hardest

Paper (short title)	Year	Category	Tasks/settings	Model focus	Prompt intervention types studied	Key findings prompt-ablation X explainability
"How Sensitive Are LMMs to Prompts?"	2025	MLLM+prompt performance (direct)	MCQA across multiple multimodal benchmarks	open vs proprietary LMMs	prompt types incl. positive/negative/neutral styles; "focus only on image/question"	Prompt phrasing significantly alters performance; derives prompting principles
EAGLE (token reliance explanations)	2026	MLLM explainability (direct)	attribution + hallucination diagnosis	open-source MLLMs	not primarily prompt-focused	Explains w/ regions support tokens; disentangling language priors vs perceptual evidence
LLaVA-CAM + information flow	2025	MLLM+explainability (direct)	VQA + captioning; truncation studies	LVLMS incl. LLaVA/ InternVL/ Qwen	prompt-dependent "cliff layer" effect	Prompt/context affects where image tokens matter; gradient-based visualization for token contribution
Leading questions robustness benchmark (MMR)	2024	prompt brittleness (direct)	MCQA w/ misleading prompts	MLLMs	paired positive vs misleading negative questions	Quantifies "misleading rate" and robustness accuracy under prompt interference
Prompt Highlighter	2024	prompting/control (direct)	generation control	LLM + VLM	token-span highlighting; conditional/unconditional context pairing	Training-free token-level control; reduces reliance on elaborate prompt engineering

Paper (short title)	Year	Category	Tasks/settings	Model focus	Prompt intervention types studied	Key findings on prompt-ablation X explainability
ControlMLLM (training-free visual prompt learning)	2024	prompting/control (direct)	region referring, OCR-ish, localization prompts	MLLMs	visual prompts (box/mask/point/scribble) + text prompts	Inference-optimization of visual tokens to steer attention; interpretability claims
Multi-Scale + ARC intra-modal attribution	2025	explainability (direct)	captioning + grounding datasets	LLaVA/ Qwen2-VL/ InternVL2.5	(implicit) context-token interference mitigation	Uses multi-scale inputs to suppress spurious prompt token activations for cleaner attribution
OPERA	2024	hallucination + attention analysis	open-ended generation	multiple LVLMs	decoding-time penalties; relates to prompt tokens/special tokens	Hallucinations correlate with attention aggregation patterns on low-information tokens
VDGD (Visual Description Grounded Decoding)	2025	prompting/decoding (indirect)	reasoning vs recognition prompts	LVLMs (e.g., LLaVA)	adds detailed description prefix; studies rephrased prompts	“Cognitive prompts” remain helpful after rephrasing; prompts changes outcomes; prefix grounding helps
CLIP Surgery	2023/2024	methodology (indirect)	open-vocab + explainability	CLIP	not prompt, but explanation reliability	Diagnoses noisy/opportunistic CAMs; proposes inference “surgery” for better maps

Paper (short title)	Year	Category	Tasks/settings	Model focus	Prompt intervention types studied	Key findings on prompt-ablation X explainability
Vision Transformers Need Registers	2024	methodology (indirect)	ViT interpretability	ViTs	none	Register tokens reduce attention artifacts; relevant to attention-interpretability limits
Brittlebench (LLM prompt robustness)	2026	methodology (LLM-only)	text benchmarks	LLMs	semantics-preserving perturbations taxonomy	Prompt perturbations can explain major variance; variance decomposition method transferable
PromptBench	2024	methodology (LLM-only)	text benchmarks	LLMs	integrated prompt attacks/perturbations	Toolkit for prompt robustness evaluation (transferal design)

Annotated entries

Directly relevant: MLLM explainability methods with token-level visual grounding or token-conditioned attribution

Token Activation Map to Visually Explain Multimodal LLMs (TAM) (2025) — first author: Yi Li ⁹ et al. Peer-reviewed (ICCV).

Category: directly relevant (MLLM+prompt+explainability). **Flags:** TAM/TAM-like; context-token interference; token-level activation maps.

Summary: TAM targets a core MLLM-specific interpretability problem: because MLLMs generate tokens progressively, **earlier context tokens (prompt + prior answer tokens)** can introduce **redundant activations** that contaminate the explanation of later tokens; TAM proposes an estimated causal-inference correction plus a rank Gaussian filter to remove context interference and denoise activation maps. ¹⁰

Evaluation/metrics: argues for **plausibility over perturbation-based faithfulness** in MLLMs (perturbations can change generation and invalidate the test); defines **Obj-IoU**, **Func-IoU**, **F1-IoU** and reports quantitative improvements for Qwen2-VL-2B and transfer across Qwen2-VL, InternVL2.5, and LLaVA-1.5. ¹¹

Why relevant: this is the anchor paper for any prompt-ablation study on TAM-like explanations: it formalizes *context interference* and provides concrete, token-level visual explanation metrics for MLLMs.

¹²

Inform: both (output quality + explanation quality), with primary emphasis on explanation quality.

Target model families: Qwen2-VL / InternVL2.5 / LLaVA-1.5. ¹³

Code: available. ¹⁴

Confidence: high.

Where MLLMs Attend and What They Rely On: Explaining Autoregressive Token Generation

(EAGLE) (2026) — first author: Ruoyu Chen ¹⁵ et al. Accepted (CVPR 2026; arXiv).

Category: directly relevant (MLLM+explainability). **Flags:** token-level attribution; modality reliance; faithfulness/localization claims.

Summary: EAGLE is a black-box explanation framework that attributes chosen generated tokens to compact image regions while also quantifying **relative influence of language priors vs perceptual evidence**, with sufficiency/necessity-style scores optimized via greedy search over sparsified regions.

¹⁶

Why relevant: valuable as (i) a complementary token-attribution baseline to TAM, and (ii) a way to test whether prompt interventions shift tokens from “vision-reliant” to “language-prior-reliant,” which is critical for stress-testing explanation stability and hallucination diagnosis. ¹⁶

Inform: explainability (primary) and output diagnosis (secondary).

Target model families: “open-source MLLMs” broadly; usable with your target families as long as interfaces exist. ¹⁶

Code: available. ¹⁷

Confidence: high (paper + repo).

Explaining multimodal LLMs via intra-modal token interactions (MSEA + ARC) (2025) — first author: Jiawei Liang ¹⁸ et al. Preprint (arXiv); also appeared as a withdrawn submission on OpenReview (ICLR 2026).

Category: directly relevant (MLLM+explainability). **Flags:** context-token interference mitigation; multi-scale attribution; TAM-adjacent.

Summary: argues that cross-modal attribution alone is insufficient: (a) isolated visual tokens have limited receptive fields, producing fragmented attribution; (b) prior text tokens induce spurious activations. Proposes **Multi-Scale Explanation Aggregation (MSEA)** by running multiple input scales and fusing attribution maps; for textual interference, proposes **Activation Ranking Correlation (ARC)** to down-weight/subtract irrelevant prior-token influence. ¹⁹

Why relevant: this is one of the closest conceptual neighbors to TAM for your hypothesis: *prompt ablations that change context-token similarity* should measurably change ARC-like signals and explanation noise/stability. ²⁰

Inform: explainability (primary).

Target model families: explicitly lists LLaVA-1.5, Qwen2-VL, InternVL2.5 in experiments. ²¹

Code: not confirmed in accessible sources here (screen during full-text review). ²²

Confidence: medium (strong conceptual fit; venue status unclear due to withdrawal).

From Redundancy to Relevance: Information Flow in LVLMs Across Reasoning Tasks (introducing LLaVA-CAM) (2025) — first author: Xiaofeng Zhang ²³ et al. Peer-reviewed (NAACL 2025).

Category: directly relevant (MLLM+explainability). **Flags:** LLaVA-CAM; prompt-dependent information-flow “cliff layer”; token contribution visualization.

Summary: proposes combining attention-score analysis (forward) with a gradient-based method (LLaVA-CAM, backward) to analyze how system/prompt/image tokens influence answer tokens through layers; reports a consistent pattern of **information flow convergence in shallow layers** and **prompt-dependent “cliff layers”** where image tokens contribute little, validated via truncation experiments.

²⁴

Why relevant: prompt ablations are not just “surface changes”—this work suggests prompts can change **where** (which layers) the model relies on image vs text, which should directly impact TAM-like

maps and their stability. ²⁵

Inform: explainability and output behavior (both).

Target model families: includes LLaVA/InternVL/Qwen families in analysis claims. ²⁵

Code: available. ²⁵

Confidence: high.

OPERA: Alleviating Hallucination in Multi-Modal Large Language Models via Over-Trust Penalty and Retrospection-Allocation (2024) — first author: Qidong Huang ²⁶ et al. Peer-reviewed (CVPR 2024).

Category: relevant methodology (hallucination + attention analysis). **Flags:** prompt/special-token effects; attention aggregation patterns.

Summary: finds hallucinations correlate with recurring self-attention “aggregation patterns” (e.g., attention concentrating on low-information tokens like punctuation), then proposes decoding-time penalties and a retrospection-allocation strategy to mitigate hallucination without extra training. ²⁷

Why relevant: for prompt ablation, OPERA motivates a concrete hypothesis: prompt templates that induce more “summary-token” behavior (or engage certain special tokens) may increase hallucination and degrade explanation plausibility/stability; it also supplies attention-based diagnostics to relate with TAM outputs. ²⁸

Inform: both (output quality via hallucination reduction; interpretability diagnostics via attention).

Target model families: multiple LVLMS (e.g., LLaVA-1.5 referenced in paper figures/analysis). ²⁹

Code: available. ³⁰

Confidence: high.

ControlMLLM: Training-Free Visual Prompt Learning for Multimodal Large Language Models (2024) — first author: Mingrui Wu ³¹ et al. Peer-reviewed (NeurIPS 2024).

Category: directly relevant (prompting/control affecting grounding). **Flags:** visual prompting; attention steering; interpretability claim.

Summary: introduces a training-free method to inject **visual prompts** (box/mask/scribble/point) by optimizing a latent variable at inference time to strengthen text-token attention to target visual regions; positions the approach as enabling region description/reasoning and claims out-of-domain generalization and interpretability. ³²

Why relevant: while not a pure “text prompt ablation” paper, it provides a strong **prompt-intervention mechanism** that will likely interact with TAM-style explanations (visual prompts alter attention pathways, potentially changing context interference and token grounding). ³²

Inform: both (output and grounding/explainability).

Target model families: MLLMs broadly; practical for LLaVA-style architectures. ³²

Code: available. ³²

Confidence: high.

A Concept-Based Explainability Framework for Large Multimodal Models (2024) — first author: Jash Parekh ³³ et al. Peer-reviewed (NeurIPS 2024).

Category: relevant methodology (explainability). **Flags:** concept-level interpretability for multimodal inputs.

Summary: proposes concept-based approaches (concept activation vector style) to decode internal representations under multimodal inputs, aiming for more human-aligned explanation units than raw saliency. ³⁴

Why relevant: prompt ablation studies often face the “plausible-but-not-faithful” risk; concept-based methods can serve as complementary probes for whether prompt edits change internal semantic representation consistency, not only pixel-level attribution. ³⁴

Inform: explainability.

Target model families: general multimodal models. ³⁴

Code: unknown (screen). ³⁴

Confidence: medium.

Prompt sensitivity and multimodal robustness benchmarks

PARC: A Quantitative Framework Uncovering the Symmetries within Vision Language Models (2025) — first author: Jonas Schmalzfuss ³⁵ et al. Peer-reviewed (CVPR 2025).

Category: directly relevant (MLLM+prompt+performance). **Flags:** prompt variations taxonomy; negation sensitivity; calibration for comparability; explicitly highlights InternVL2 reliability.

Summary: defines realistic language+vision prompt variations and introduces a calibrated reliability score to compare prompt sensitivity across prompts/datasets; reports VLMs struggle particularly with **semantic changes such as negations** and that model families differ in reliability (with InternVL2 family highlighted as more reliable in experiments). ³⁶

Why relevant: PARC gives you an immediate blueprint for **prompt-ablation factor design** (reformulations vs semantic changes; language vs vision variations), plus a caution: comparisons across prompt variants require accounting for expected-random-performance shifts (especially when prompt changes alter the correct answer distribution). ³⁷

Inform: output quality (primary); explanation stability indirectly (by defining controlled prompt perturbations).

Target model families: includes InternVL2; methodology applies broadly. ³⁶

Code/data: stated as publicly released in the paper (verify repo link during screening). ³⁷

Confidence: high.

How Sensitive Are Large Multimodal Models to Prompts? (2025) — first author: M. I. Ismithdeen ³⁸ et al. Preprint/OpenReview PDF.

Category: directly relevant (MLLM+prompt+performance). **Flags:** prompt types and prompting principles; open vs proprietary comparison.

Summary: performs a broad prompt sensitivity analysis on multimodal multiple-choice benchmarks and distills “prompting principles,” including findings that some explicit “focus only on image/question” instructions can hurt proprietary models, and that answer leakage degrades performance. ³⁹

Why relevant: this is a direct source of prompt manipulation dimensions and observed variance; it can seed your ablation axes (positive/negative/neutral prompt types; instruction specificity; “avoid bias” instructions; focus directives). ³⁹

Inform: output quality primarily; explanation quality indirectly through changed generations and context.

Target model families: multiple “open-source and proprietary LLMs.” ³⁹

Code: not confirmed in accessible lines here (screen). ³⁹

Confidence: medium-high.

Seeing Clearly, Answering Incorrectly: A Multimodal Robustness Benchmark for Evaluating MLLMs on Leading Questions (2024) — first author not extracted in-line here (arXiv HTML). Preprint.

Category: directly relevant (prompt brittleness in multimodal). **Flags:** misleading prompts; negative questions; explicit robustness metrics.

Summary: constructs paired positive (understanding) vs leading negative (misleading) questions at character/attribute/context levels; introduces “Misleading Rate” and “Robustness Accuracy” to quantify how misleading prompts derail otherwise correct understanding. ⁴⁰

Why relevant: an ideal testbed for prompt ablation: the benchmark operationalizes **prompt interference** explicitly, which you can pair with explanation robustness (does the explanation remain image-grounded when prompts are misleading?). ⁴⁰

Inform: both (output robustness and grounding robustness).

Target model families: MLLMs broadly. ⁴⁰

Code: unknown (screen). ⁴⁰

Confidence: medium.

MVTamperBench: Evaluating Robustness of Vision-Language Models (2024/2025) — first author: Amit Agarwal ⁴¹ et al. Preprint (arXiv), plus later venue mentions.

Category: benchmark/dataset (robustness). **Flags:** video; systematic perturbations (rotation/masking/substitution/repetition/dropping).

Summary: evaluates MLLM robustness under multiple video tampering techniques across tasks, showing wide variability and that model size does not guarantee robustness. ⁴²

Why relevant: although not “prompt ablation” per se, it supports the broader stress-testing methodology: **controlled input perturbations** plus multi-task evaluation; it can inspire analogous *prompt perturbation suites* and stability metrics for explanations in multi-frame contexts. ⁴³

Informs: output robustness; explanation robustness indirectly (through stability design patterns).

Target model families: many video MLLMs. ⁴⁴

Code/data: released (project + dataset links in arXiv abstract). ⁴⁵

Confidence: high.

V2R-Bench: Holistically Evaluating LVLm Robustness to Fundamental Visual Variations (2025) — first author: Zhe Fan ⁴⁶ et al. Peer-reviewed (ACL Findings) / arXiv.

Category: benchmark/dataset (robustness, stability metrics). **Flags:** token-level stability; performance consistency; LLM-as-a-judge mentioned.

Summary: proposes a framework for robustness to fundamental visual variations and includes robustness metrics such as **performance consistency** and **semantic/token-level stability**. ⁴⁷

Why relevant: directly aligns with your emphasis on explanation stability: even when the perturbation is visual, the framework emphasizes **token-level stability**, which can be adapted to prompt perturbations and explanation-map stability. ⁴⁸

Informs: output stability; explanation stability by methodological transfer.

Target model families: LVLms broadly. ⁴⁹

Code: not confirmed in accessible lines (screen). ⁵⁰

Confidence: medium-high.

VLM-RobustBench: A Comprehensive Benchmark for Robustness of Vision-Language Models (2026) — first author: Rohit Saxena ⁵¹ et al. Preprint (arXiv).

Category: benchmark/dataset (robustness). **Flags:** perturbations; compares model families including Qwen/InternVL.

Summary: evaluates robustness under a range of perturbations; reports that spatial/geometric distortions can degrade performance strongly and that current VLMs can be semantically strong but spatially fragile. ⁵²

Why relevant: complements prompt ablation with a reminder: if model behavior is highly variant under *input distortions*, explanation stability should be tested under both prompt and visual perturbations to isolate their contributions. ⁵²

Informs: output robustness primarily.

Target model families: includes Qwen and InternVL families (per abstract). ⁵²

Code: unknown (screen). ⁵²

Confidence: medium.

Prompting methods that change reasoning, compositionality, and controllability

Compositional Chain-of-Thought Prompting for Large Multimodal Models (CCoT) (2024) — first author: Chandan Mitra ⁵³ et al. Peer-reviewed (CVPR 2024).

Category: directly relevant (MLLM+prompt+performance). **Flags:** decomposition; structured prompting;

intermediate representations.

Summary: proposes a structured, two-step CoT that first generates a **scene graph** conditioned on the task prompt, then answers using that intermediate; aims to improve compositional reasoning without finetuning (prompting-only). ⁵⁴

Why relevant: prompt ablation should explicitly include **decomposition prompts** and intermediate-structure prompts, because they change generation length, token distribution, and reliance on visual evidence (all of which can shift TAM-style token attribution maps). ⁵⁴

Inform: output quality and explainability plausibility (indirectly, by controlling intermediate reasoning).

Target model families: large multimodal models broadly. ⁵⁴

Code: not confirmed here (screen). ⁵⁴

Confidence: high.

Visual CoT: Advancing Multi-Modal Language Models with a Comprehensive Dataset and Benchmark for Chain-of-Thought Reasoning (2024) — first author: Hao Shao ⁵⁵ et al. Peer-reviewed (NeurIPS 2024 Datasets & Benchmarks).

Category: benchmark/dataset (prompting + grounding). **Flags:** visual CoT; intermediate bounding boxes; interpretability hooks.

Summary: introduces a large dataset with intermediate bounding boxes and reasoning steps, and a multi-turn pipeline to focus on visual inputs while providing interpretable “thoughts.” ⁵⁶

Why relevant: strong for ablation: you can test whether prompt changes (CoT vs no-CoT; different rationales) modify not only answer accuracy but also the spatial grounding (boxes) and token-level explanation stability. ⁵⁷

Inform: both.

Target model families: MLLMs broadly. ⁵⁸

Code: available. ⁵⁹

Confidence: high.

Multimodal Chain-of-Thought Reasoning in Language Models (Multimodal-CoT) (2023) — first author: Zehui Zhang ⁶⁰ et al. OpenReview PDF (peer-reviewed status depends on venue; treat as preprint unless confirmed during screening).

Category: relevant methodology (CoT in multimodal). **Flags:** two-stage rationale then answer; hallucination discussion.

Summary: advocates a two-stage multimodal CoT framework separating rationale generation and answer inference; discusses limitations of converting vision into captions and information loss, and evaluates on multimodal reasoning datasets. ⁶¹

Why relevant: for your study, it gives canonical CoT prompting knobs (rationale length, structure, separation of phases) that likely influence token-level explanations and their stability. ⁶²

Inform: output quality primarily; explainability indirectly.

Target model families: multimodal reasoning models generally. ⁶²

Code: unknown here (screen). ⁶²

Confidence: medium.

Prompt Highlighter: Interactive Control for Multi-Modal LLMs (2024) — first author: Yuechen Zhang ⁶³ et al. Peer-reviewed (CVPR 2024).

Category: directly relevant (prompt intervention). **Flags:** token-level prompt interactions; system-like control without rewriting prompts.

Summary: proposes a training-free inference method enabling users to highlight spans of the prompt and guide generation via attention-weight steering; motivated by classifier-free guidance; claims more controllable outputs than vanilla inference or prompt engineering. ⁶⁴

Why relevant: provides a concrete “prompt ablation operator” (highlight vs not; highlight different spans) that changes token generation while keeping surface prompt nearly fixed—ideal for isolating

explanation stability effects. ⁶⁵

Informs: both (output and interpretability/control).

Target model families: compatible with LLMs and VLMs; demonstrated on LLaVA-v1.5. ⁶⁶

Code: available. ⁶⁷

Confidence: high.

Attention Prompting on Image (API) for Large Vision-Language Models (2024) — first author: Runpeng Yu ⁶⁸ et al. Peer-reviewed (ECCV 2024).

Category: directly relevant (prompt intervention via vision). **Flags:** query-conditioned visual prompts; attention heatmaps.

Summary: constructs a query-dependent attention heatmap (e.g., with CLIP), overlays/multiplies it into the image to guide LLaVA attention; reports improvements on benchmarks for LLaVA-1.5, and positions visual prompting as a training-free alternative to finetuning. ⁶⁹

Why relevant: although not text-only, it adds an important control arm: **visual prompt ablations** may stabilize grounding and thus stabilize TAM-like explanations (or reveal mismatches between explanation and actual evidence). ⁶⁹

Informs: both.

Target model families: explicitly reports improvements for LLaVA-1.5. ⁶⁹

Code: not confirmed here (screen). ⁶⁹

Confidence: high.

Surveys, datasets, and toolkits for building a complete state-of-the-art picture

Visual Prompting in Multimodal Large Language Models (survey) (2024) — first author: Jiaxi Wu ⁷⁰ et al. Preprint (arXiv).

Category: survey/review.

Summary: a survey focused specifically on “visual prompting” for MLLMs, including prompt generation, compositional reasoning, and prompt learning. ⁷¹

Why relevant: gives you high-recall coverage of visual prompting techniques to include as ablation arms (especially if you later expand beyond text prompts). ⁷¹

Informs: both (by cataloging methods).

Target model families: MLLMs broadly. ⁷¹

Code: N/A (survey).

Confidence: medium-high.

Multimodal Chain-of-Thought Reasoning (survey) (2025) — first author: Yaoting Wang ⁷² et al. Preprint (arXiv).

Category: survey/review.

Summary: positions itself as a systematic survey of MCoT reasoning with taxonomy and analysis across application scenarios. ⁷³

Why relevant: provides a structured map of CoT-style prompt interventions in multimodal settings—high value for recall and consistent naming of ablation factors. ⁷³

Informs: output quality primarily; explainability indirectly.

Confidence: medium.

VLMEvalKit: An Open-Source Toolkit for Evaluating Large Multi-Modality Models (2024–2025) — first author: Haodong Duan ⁷⁴ et al. Preprint (arXiv) + actively maintained code.

Category: benchmark/dataset/code toolkit.

Summary: provides a unified interface to evaluate many multimodal models and benchmarks; reports large and growing coverage (models/benchmarks) across versions, and includes generation-based evaluation and answer extraction options. ⁷⁵

Why relevant: essential infrastructure for prompt ablation at scale—lets you run many prompt variants across many benchmarks reproducibly, and standardizes pre/post-processing. ⁷⁶

Informs: both (enables output evaluation; can be extended to explanation evaluation pipelines).

Code: available. ⁷⁷

Confidence: high.

Hallucination of Multimodal Large Language Models: A Survey (2024) — first author: Zechen Bai ⁷⁸
et al. Preprint (arXiv) + resources repo.

Category: survey/review + resource list.

Summary: surveys hallucination in MLLMs/LVLMs: causes, evaluation benchmarks, mitigation; links to a curated resource repository. ⁷⁹

Why relevant: hallucination is often the first observable symptom of “prompt over-reliance,” and hallucination metrics/benchmarks (POPE, etc.) can serve as downstream correlates when explanation stability changes under prompt ablation. ⁷⁹

Informs: output quality primarily; explainability indirectly.

Confidence: medium-high.

A Survey on Hallucination in Large Vision-Language Models (2024) — first author: Hanchao Liu ⁸⁰
et al. Preprint (arXiv).

Category: survey/review.

Summary: focuses on LVLM hallucination symptoms, benchmarks, root causes, and mitigations. ⁸¹

Why relevant: again, gives broad benchmark coverage; helps ensure your prompt ablation suite includes hallucination-focused tasks where grounding/explanation divergence is likely. ⁸¹

Informs: output quality.

Confidence: medium-high.

POPE: Polling-based Object Probing Evaluation for Object Hallucination (2023) — repo documents “Evaluating Object Hallucination in Large Vision-Language Models (EMNLP 2023)”.

Category: benchmark/dataset/code. **Flags:** widely used hallucination metric; prompt-template ablations in follow-up work often report POPE.

Summary: provides code/data for POPE, a probing evaluation of object hallucination; supports building POPE variants from segmentation tools and datasets. ⁸²

Why relevant: an essential evaluation axis: prompt changes that increase language priors often increase hallucinated objects, and explanation maps may drift from image evidence; POPE gives a measurable downstream endpoint. ⁸³

Informs: output quality; also explanation validation indirectly.

Code/data: available. ⁸³

Confidence: high.

Indirect but methodologically important foundations for TAM-like explanation robustness

A Closer Look at the Explainability of Contrastive Language-Image Pre-training (CLIP Surgery) (2023; revised 2024) — first author: Yi Li et al. Preprint (arXiv).

Category: relevant methodology (CLIP explainability). **Flags:** noisy/opposite CAM; inference-time modification.

Summary: diagnoses two explainability issues in CLIP: background focus (“opposite visualization”) attributed to inconsistent semantic regions in raw self-attention, and noisy activations due to redundant features; proposes “CLIP Surgery” inference modifications to improve CAM reliability and open-vocabulary capacity; code linked. ⁸⁴

Why relevant: TAM explicitly compares against CLIP/Vit explainability baselines and inherits CAM-family thinking; CLIP Surgery highlights that even in contrastive models, attribution maps can be misleading

without careful treatment—useful as a cautionary baseline when interpreting TAM-like maps across model families. ⁸⁵

Inform: explainability.

Code: available (linked on arXiv page). ⁸⁶

Confidence: medium-high.

Vision Transformers Need Registers (2024) — first author: Timothée Darcet ⁸⁷ et al. Peer-reviewed (ICLR 2024).

Category: relevant methodology (ViT interpretability).

Summary: identifies artifacts in ViT feature maps/attention; proposes adding register tokens to store global info and reduce artifacts, improving dense-task behavior and interpretability of attention maps. ⁸⁸

Why relevant: TAM-like methods often depend on token/patch-level mappings; this paper (and follow-ups) warn that attention maps can contain artifacts and that architectural/token choices affect interpretability. ⁸⁹

Inform: explainability methodology.

Code: implementations exist; verify official artifacts in screening. ⁹⁰

Confidence: high.

Register and [CLS] tokens induce a decoupling of local and global features in large ViTs (2025) — first author: Alexander Lappe ⁹¹ et al. Peer-reviewed (NeurIPS 2025).

Category: relevant methodology (limitations of attention-map interpretability).

Summary: examines how register/[CLS] tokens can break the “patch integration assumption,” showing that even “cleaner” attention maps may not faithfully reflect how local patch evidence contributes to global outputs, especially in large models. ⁹²

Why relevant: strengthens the argument that **explanation faithfulness/stability cannot be assumed from attention visualization alone**, supporting TAM’s emphasis on dedicated plausibility metrics and context-interference correction. ⁹³

Inform: explainability methodology.

Confidence: medium-high.

Brittlebench: Quantifying LLM Robustness via Prompt Sensitivity (2026) — first author: Alex Wu ⁹⁴ (and coauthors; see paper). Preprint (arXiv).

Category: relevant methodology (LLM prompt robustness). **Flags:** variance decomposition; semantics-preserving perturbations; ranking instability.

Summary: proposes a theoretical framework and evaluation pipeline for prompt brittleness using semantics-preserving perturbations; reports that prompt-induced variability can be large and can change model rankings frequently; attributes up to ~half of performance variance to prompt perturbations for a model in some settings. ⁹⁵

Why relevant: while text-only, Brittlebench contributes a transferable statistical idea: **variance decomposition** separating difficulty vs prompt sensitivity—directly applicable to separating “output drift” vs “explanation drift” components under prompt ablation. ⁹⁵

Inform: methodology for robustness; indirect for explainability stability.

Code: “to be released” per paper lines shown. ⁹⁵

Confidence: medium.

A Unified Library for Evaluation of Large Language Models (PromptBench) (2024) — first author: Kai Zhu ⁹⁶ et al. Peer-reviewed (JMLR).

Category: relevant methodology (LLM prompt robustness evaluation toolkit).

Summary: provides an evaluation library that includes adversarial prompt attacks/perturbations to study robustness on prompts. ⁹⁷

Why relevant: useful as an engineering reference for how to implement and organize prompt perturbation operators (char/word-level edits, paraphrases, etc.) even if your target is an MLLM. ⁹⁷

Informs: methodology/tooling.

Confidence: medium.

Separate list of explicitly non-peer-reviewed resources (useful, but treat as weak evidence)

The following are **non-academic** or **curation repos**; they can improve recall and help discover papers/code, but should not be treated as evidence for claims without following the primary sources.

- Curated list: GitHub ⁹⁸ repo “Awesome-LMMs-Mechanistic-Interpretability” (good discovery surface for mechanistic interpretability in multimodal). ⁹⁹
- Curated list: “Awesome Multimodal LLMs” repository (broad discovery surface; requires careful verification). ¹⁰⁰
- Curated list: “Awesome LVLM Hallucination” repository (paper discovery on hallucination/robustness). ¹⁰¹
- Blog-style evaluation write-ups (screen carefully; may be useful for benchmark taxonomy but not evidence): multimodal benchmark explainer page. ¹⁰²
- Community threads/issues sometimes documenting prompt-template sensitivities (treat as anecdotal unless confirmed): LLaVA issue mentioning LLaVA-CAM code release plans. ¹⁰³
- Third-party summaries (Moonlight, TLDR, etc.)—use only to locate papers quickly, then verify from primary PDF. ¹⁰⁴

Thematic synthesis and gaps for prompt ablation on TAM-like explanations

Thematic clusters

```
flowchart TB
    A[Prompt interventions] --> B[Output changes]
    A --> C[Context-token distribution & similarity]
    C --> D[Token-level explanation interference]
    D --> E[TAM-style causal correction]
    D --> F[ARC-style context filtering]
    B --> G[Hallucination & grounding failures]
    G --> H[Benchmarks: POPE / leading questions / robustness suites]
    E --> I[Plausibility metrics: Obj-IoU / Func-IoU / F1-IoU]
    F --> I
    J[Prompt sensitivity frameworks] --> A
    J --> H
    K[Control & visual prompting] --> A
    K --> D
```

This diagram reflects a synthesized view: prompt interventions influence outputs and also reshape the **token context** that drives interference in token-level explanations; TAM-like methods (and related ARC-style mitigation) are designed to remove or reduce that interference. The evaluation loop closes through plausibility/grounding/hallucination benchmarks. ¹⁰⁵

Timeline of key developments

```
timeline
  title Key milestones for prompt robustness and token-level multimodal explainability
  2023 : POPE (object hallucination eval) : CLIP Surgery (CLIP explainability diagnosis)
  2024 : OPERA (hallucination + attention patterns) : Prompt Highlighter (token-level prompt control)
  2024 : ControlMLLM (training-free visual prompting) : PARC (prompt sensitivity framework)
  2025 : LLaVA-CAM + info-flow analysis : TAM (token activation maps for MLLMs)
  2025 : Leading-questions robustness benchmarks : V2R-Bench (stability metrics)
  2026 : EAGLE (token reliance, priors vs vision) : Brittlebench (prompt robustness meta-eval)
```

(Entries align to the primary sources cited in the corpus.) ¹⁰⁶

High-value transferable methods for your prompt-ablation study

A robust prompt-ablation study on TAM-like explainability will likely benefit from combining these ideas:

- 1) **Prompt variation taxonomies that separate reformulation vs semantic change** (especially negation and answer-space changes). PARC provides explicit motivation and calibration for comparability across prompt variants. ³⁶
- 2) **Prompt sensitivity evaluation across multimodal benchmarks** (open vs proprietary; multiple prompt “types”). “How Sensitive Are Large Multimodal Models to Prompts?” provides prompt-type principles and empirically observed sensitivities. ³⁹
- 3) **Token-level explanation interference modeling**: TAM frames interference as context-token redundancy contaminating later-token activation maps and proposes causal correction. ¹⁰
- 4) **Alternative explanation frameworks** to cross-check TAM results: EAGLE’s priors-vs-vision reliance split and sufficiency/necessity framing can validate whether prompt changes shift the “source of evidence.” ¹⁶
- 5) **Stability metrics beyond raw accuracy**: leading-question benchmarks define explicit robustness measures under prompt interference; V2R-Bench emphasizes token-level stability/consistency concepts adaptable to prompt variations. ¹⁰⁷
- 6) **Cautionary lessons from attention-map interpretability in ViTs/CLIP** (register tokens, decoupling), suggesting that attention visualizations can be artifact-prone and motivating metric-based plausibility checks. ¹⁰⁸

Open gaps specific to prompt ablation on TAM-like token activation maps

The literature still leaves several “project-grade” gaps—these can become your contribution if you execute prompt ablation rigorously:

- **Prompt → explanation stability as a first-class metric**: many works measure prompt sensitivity on accuracy, but fewer quantify how explanations (token maps) drift under semantics-

preserving rewordings. This is precisely where you can extend PARC/Brittlebench-style perturbations to explanation-space variance decomposition. ¹⁰⁹

- **System vs user prompt effects on grounding/explanations:** sensitivity studies rarely isolate system-message constraints vs user-message content in multimodal settings; this is an actionable ablation axis for your tools (ChatGPT Deep Research vs Perplexity vs Consensus usage also differs). ¹¹⁰
- **Chain-of-thought / decomposition prompts vs token-level grounding:** CoT improves performance in many settings, but whether it improves *faithful grounding* (and stable token maps) is unclear; existing work provides datasets and prompting methods, but not a unified explanation-stability analysis. ¹¹¹
- **Multi-turn dialogue and long outputs:** TAM discusses multi-turn and multi-image/video applicability, but systematic prompt-ablation protocols for long contexts (where interference should grow) are not standardized. ¹¹²
- **Evaluation choice (plausibility vs faithfulness):** TAM argues against perturbation-based faithfulness for MLLMs due to generation instability; your work can concretize this by comparing plausibility metrics (Obj/Func/F1-IoU) with carefully designed “faithfulness-like” tests that keep generation fixed (e.g., logit conditioning or constrained decoding), if feasible. ¹¹³

Engine-optimized search query pack

Below is a **high-recall, structured query pack** designed for use across (i) ChatGPT Deep Research, (ii) Perplexity, and (iii) Consensus. It is tailored to your project’s intersection: **prompt interventions × MLLM explainability/grounding × TAM-style token-level maps**. Replace bracketed terms with your exact model/dataset names when needed.

Master “Deep Research” prompt template (copy/paste)

Use this as a single orchestration prompt; then run follow-up queries per cluster.

You are conducting a high-recall structured literature reconnaissance (2023–2026 prioritized, older foundational allowed) on:

(1) How prompt interventions affect Multimodal Large Language Models (MLLMs/ LVLMS/VLMs) outputs,
(2) How prompt interventions affect token-level visual explanations/grounding quality, stability, plausibility/faithfulness,
with emphasis on TAM-style token activation maps (token-conditioned activation/attribution maps for generated tokens).

Models of interest (priority): Qwen2-VL, LLaVA-1.5, InternVL2.5. Include other MLLMs when relevant.

Search objectives (high recall):

A. Papers that directly study prompt sensitivity/variance in MLLMs (prompt paraphrasing, negation, format, system vs user, role prompting, CoT, decomposition, self-refinement, structured output).

B. Papers that provide MLLM visual explainability methods (token-level grounding, activation maps, saliency/Grad-CAM variants, attention/attribution, causal interventions, probing) and discuss explanation reliability/stability.

C. Works that connect prompts to hallucination/grounding failures, or propose prompt-based mitigation.

D. Surveys/reviews, benchmarks/datasets, toolkits, and code repos useful for running prompt ablations and evaluating explanation plausibility/stability.

For each retrieved item, extract:

- title, year, authors (first author + et al.), venue/source type (peer-reviewed / preprint / non-academic),
- category (MLLM+prompt+explainability / MLLM+prompt+performance / methodology / survey / benchmark-dataset-code / non-peer),
- tasks/settings (captioning, VQA, visual reasoning, localization/grounding, dialogue, multi-image/video),
- prompt interventions tested (explicitly list), and whether interventions are semantics-preserving vs semantics-changing,
- explainability method and what tokens/regions it explains; whether it is TAM-like,
- metrics used (e.g., Obj-IoU / Func-IoU / F1-IoU; stability/consistency; hallucination metrics like POPE),
- datasets used, model family, code availability,
- limitations and whether evidence is direct vs inferential.

Do not fabricate. Prefer primary sources (arXiv, CVF/ACM/ACL proceedings, official GitHub).

Output a structured report + table + clusters + timeline.

Query set for Perplexity (high recall, fast coverage)

Run these as separate searches (use Academic mode + recency filter where possible):

- 1) "token-level" visual explanation multimodal large language model prompt interference
- 2) "Token Activation Map" TAM multimodal LLM context token interference
- 3) prompt sensitivity "vision language model" negation paraphrase benchmark
- 4) system prompt vs user prompt robustness multimodal
- 5) LLaVA-CAM gradient-based explanation image token contribution
- 6) EAGLE explain autoregressive token generation multimodal
- 7) Visual CoT dataset bounding boxes reasoning steps prompt
- 8) Compositional chain-of-thought prompting large multimodal models scene graph
- 9) leading questions multimodal robustness benchmark misleading prompts
- 10) prompt engineering hallucination mitigation large vision-language models "prompt template ablation"

These map directly to the key clusters in your corpus. ¹¹⁴

Query set for Consensus (precision-biased within high recall)

Consensus is strongest when you phrase explicit research questions:

- 1) "Do prompt paraphrases change performance in multimodal LLMs?" ¹¹⁰
- 2) "How does negation in prompts affect VLM accuracy and reliability?" ³⁷

- 3) "How do context tokens interfere with token-level visual explanations in MLLMs?" 115
- 4) "What metrics evaluate plausibility of token-level explanation maps in MLLMs?" 3
- 5) "How to detect misleading prompts / leading questions robustness in multimodal benchmarks?" 40
- 6) "Do chain-of-thought prompts improve visual grounding in multimodal models?" 111
- 7) "What toolkits exist for evaluating large vision-language models across benchmarks?" 116
- 8) "What methods explain whether MLLM tokens rely on vision vs language priors?" 16

Screening and experiment-extraction protocol

To turn this high-recall corpus into an actionable "state of the art" map for your prompt ablation work, a lightweight but rigorous protocol helps prevent wasted depth on weakly relevant papers.

Screening workflow

A practical screening sequence is:

- **Stage A: Deduplicate + classify** by (peer-reviewed / preprint / non-academic) and by the categories used above. Keep non-academic only as pointers to primary sources.
- **Stage B: Relevance tagging** with three boolean tags:
 - *Prompt-focused?* (explicit prompt variants tested)
 - *Explainability-focused?* (produces/analyses visual explanations/grounding)
 - *Token-level?* (token-conditioned maps/attribution; TAM-like)
- **Stage C: Evidence strength:** mark "direct evidence" only if the paper experimentally varies prompts and measures outcomes; mark "inferential" when prompts are not varied but the method implies prompt sensitivity. (Example: TAM gives direct context-interference claims; OPERA gives attention-pattern claims that are prompt-template-sensitive but not a prompt ablation study per se.) 117

Extraction template for each accepted paper

For your eventual internal dataset (spreadsheet/Notion), extract:

- **Prompt intervention operators:** paraphrase, negation, reorder, shorten/lengthen, role/system changes, structured output schema, CoT on/off, decomposition prompts, self-refine loop, "focus on image" directives, etc. 118
- **Which tokens are explained:** all output tokens vs selected tokens; noun tokens vs function tokens (important because TAM explicitly separates these via Obj-IoU/Func-IoU). 3
- **Explanation method details:** gradient-based vs attention-based vs model-agnostic; any context-window mitigation; any aggregation across layers/heads/scales. 119
- **Metrics:**
 - output: accuracy, robustness accuracy, misleading rate, hallucination metrics (e.g., POPE) 120
 - explainability: plausibility metrics (Obj/Func/F1-IoU), localization/faithfulness where defined 121
 - stability: define and compute consistently (e.g., rank correlation or overlap of top-k activated patches across prompt variants; variance decomposition analogue to Brittlebench). 122

Recommended experimental protocol extraction for prompt ablation on TAM-like explanations

A research-grade prompt ablation protocol, aligned with the cited literature, typically needs:

- **Prompt sets:** at minimum, separate **semantics-preserving reformulations** from **semantics-changing perturbations** (negation, swapped answers, altered constraints), because PARC shows semantic changes are particularly disruptive, and calibrated comparison is needed when the “expected answer” changes. ³⁶
- **Dual outcomes per prompt:** always record both (i) correctness/robustness and (ii) explanation plausibility/stability. This prevents the “accurate but ungrounded” vs “grounded but wrong” ambiguity that motivates multiple robustness categories in leading-question benchmarks. ¹²³
- **Token-level evaluation:** follow TAM’s separation of object vs function tokens as a template; prompt interventions often change the proportion of function tokens (formatting, verbosity), which can confound explanation metrics unless measured explicitly. ¹²⁴
- **Cross-method triangulation:** at least one TAM-like method + one orthogonal attribution approach (e.g., EAGLE or LLaVA-CAM style contribution visualization) to detect method-specific artifacts. ¹²⁵

This closes the loop between *prompt brittleness*, *output robustness*, and *token-level explanation robustness* —the exact intersection your project targets.

¹ ³ ⁴ ⁸ ⁹ ¹⁰ ¹¹ ¹² ¹³ ³³ ⁹⁴ ¹⁰⁵ ¹¹² ¹¹³ ¹¹⁴ ¹¹⁵ ¹¹⁷ ¹¹⁹ ¹²¹ ¹²⁴ ¹²⁵ https://openaccess.thecvf.com/content/ICCV2025/papers/Li_Token_Activation_Map_to_Visually_Explain_Multimodal_LLMs_ICCV_2025_paper.pdf
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